Music AI, Copyright’s Informational Remainder, and the Aggregate

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3 discussion topics (for later)

- 1) Copyright's informational remainder (CIR)
- 2) The aggregate
- 3) ML, music, and the commons – and alternative future models

Music AI, Copyright's Informational Remainder, and the Aggregate

- Copyright as a site of conflict in the music industry takes the form of a stark binary: to protect or not to protect AI training data
- But, if musicians and rights bodies were to win the battle to enforce copyright (now seemingly possible), there is a lot more to say and study
- It is essential for non-legal academics, those concerned with the cultural underpinnings of a copyright concept, to weigh in on the debate, to address 1) a broader concept of AI's impact as a cultural-economic phenomenon; 2) latent norms and ontologies that become visible only as they are unsettled by AI
- And to look towards more substantial questions of music's relationship to society, musicians' livelihoods, and possible new configurations possible in—or necessitated by––an AI music world

What does copyright leave out?

- Copyright at center of many (not all) debates over AI, fairness, and compensation
  - Lawsuits
  - Guidance from national copyright bodies
  - Legislation
- Among most vocal stakeholders in debates are individuals & entities with strong investment in copyright as an institution **on both sides**
  - Labels, publishers, unions, rights agencies, lawyers, individual artists
- But what does exclusive focus on copyright exclude?

The "AI copyright trap"

- Music copyright has evolved historically in part through response to technological change (mechanical music, sampling, file-sharing, etc.)
- Yet overreliance on copyright presents risks
- Carys Craig, "AI Copyright Trap": "the inequalities, exploitation, and private profiteering that characterize the swelling AI industry [...] are also characteristics of the copyright industries"
- Three fallacies:
  - anything of value must be rendered as private property (& thus anything that resists propertization is valueless)
  - unauthorized copying is inherently wrongful
  - copyright industries appeal to the figure of starving artists to push its further expansion

Music and / as information:
3 genealogies

1) Shannon-esque music-as-info theory

- Treatment of music as information goes back to late C19th efforts to define music as an object of scientific research
- After 1945: metaphoric mapping of music onto information gained wider prominence with development of information theory and related research areas: cybernetics, computer science, cognitive psychology, information science etc
- 1948: Shannon's foundational "A Mathematical Theory of Communication" > the paper's insights inflated and applied to the production and study of music
- EG: 1950: RCA engineers at RCA began work on a "Music Composing Machine" which drew on Shannon's statistical analyses of language aiming to automate the compositional process
- Such initiatives limited by severe restrictions placed on which sonic aspects of music could be rendered as information: generally pitch and rhythm, with dynamics, timbre and other harder-to-quantify sonic parameters largely avoided

2) Meyer, Moles and 50s ‘information aesthetics’

- Parallel developments also took place in research into music aesthetics through the work of Leonard Meyer (1957) and Abraham Moles (1958)
- While the prominence of information aesthetics receded after 1950s, its legacy persists in such fields as music cognition and corpus studies
- Here, too, the significant transformations that music has to undergo to be rendered information represents a major difficulty, giving rise to critiques like Keil’s
- For Keil (1966), Meyer’s focus on the expectations engendered by melodic-harmonic patterns overlooks the “participatory discrepancies” central to the expressive power of Afrodiasporic musics (and many other music traditions)
- By this account, informational perspectives on music sustain Eurocentric understandings of music, propagating such biased conceptualizations through their reductions

3) Music Information Retrieval (MIR)

- Third genealogy is the field of music information retrieval (MIR): offshoot of the larger (and longer-standing) fields of info science and info retrieval
- For much of its history MIR was concerned with indexing and searching of symbolic music data, i.e. notated scores
- 1990s: with large-scale digitization of audio, the centre of gravity in MIR shifted to commercial and industrial applications that have dominated the field since the early 2000s: music genre recognition, music emotion recognition, playlist generation, auto-tagging, digital fingerprinting and generative music
- In sum: all 3 genealogies share a Shannon-esque understanding of 'information': it's defined in statistical terms – an event contains more information the less probable it is, and less information the more predictable (redundant) it is
- A key technique enabled by this statistical perspective is data compression
- Yet all 3 approaches inflate Shannon's mathematical work, treating issues of predictability and probability as indexical of cultural and psychological phenomena, whereas information in Shannon's sense explicitly abjures matters of meaning

Shannon’s Information Theory

- Shannon was interested in the problem of sending signals reliably over an unreliable network. Imagine you’re sending a message and part of it gets lost on the way. It may be possible to predict the missing bits based on the context.
- Happy birthday to you, happy birthday [A] you, happy birthday dear [B], happy birthday to you. [A = confident guess from context?, B = hmm, not seen this context before].
- More predictable = less information. Less predictable = more information.
- Related to compression and entropy.
- Also related to studies of music psychology and aesthetics (see Marcus Pearce 2025, Learning to Listen, Listening to Learn: Music Perception and the Psychology of Enculturation).
- Note that pure ”noise” is 100% unpredictable, hence *high* information content. That’s why noise is a problem! Statistically, it is easily to reproduce, just not *exactly*!

Algorithmic Information Theory

- AIT specifically defines information content as: the shortest theoretically possible compression of a piece of data. (Hence "algorithmic": what is the smallest algorithm that can recreate the data?)
- Information content must be understood as mutual between pieces of data. E.g., you can compress the collected works of Shakespeare *together* more successfully than separately (one work can help predict the content in another work).
- When you compress lots of things together, some things might have more information content relative to the corpus. E.g., if the corpus consisted of all Guardian articles since 1900, and you added Hamlet, it may have more information content relative to the corpus.
- When you add something to a large compressed corpus (e.g., train a network), the *added information* can be thought of as the *thing that is unique*.

Copyright’s informational remainder (CIR)

Copyright’s Informational Remainder

- **When you add something to a large compressed corpus (e.g., train a network), the *added information* can be thought of as the *thing that is unique*.**
- For something to be copyrightable, it must be a unique expression. Whatever information was already in the corpus before that thing was added cannot be the subject of copyright. It enters the commons and is excluded *from* copyright protection.
- Our primary contention: while artists may be successfully arguing that their copyright has been infringed, there remains a huge amount of information going into generative AI that is not accounted for by copyright protection.
- This is shared cultural value that operated in a domain of free cultural exchange, but it is now turned into capital value for AI companies.

Breaking Down the Track

General acoustics of vibration Sound sticks on stretched skins Sound of tom drums Specific sound of *those* tom drums Properties of recording + mixing chain Tom drums as common fill elements Nick Mason style Specific fill pattern: - - - X X - X X
This specific recording

Not under copyright
(Most of the information)

Under copyright

Music, generative ML, and the concept of aggregate

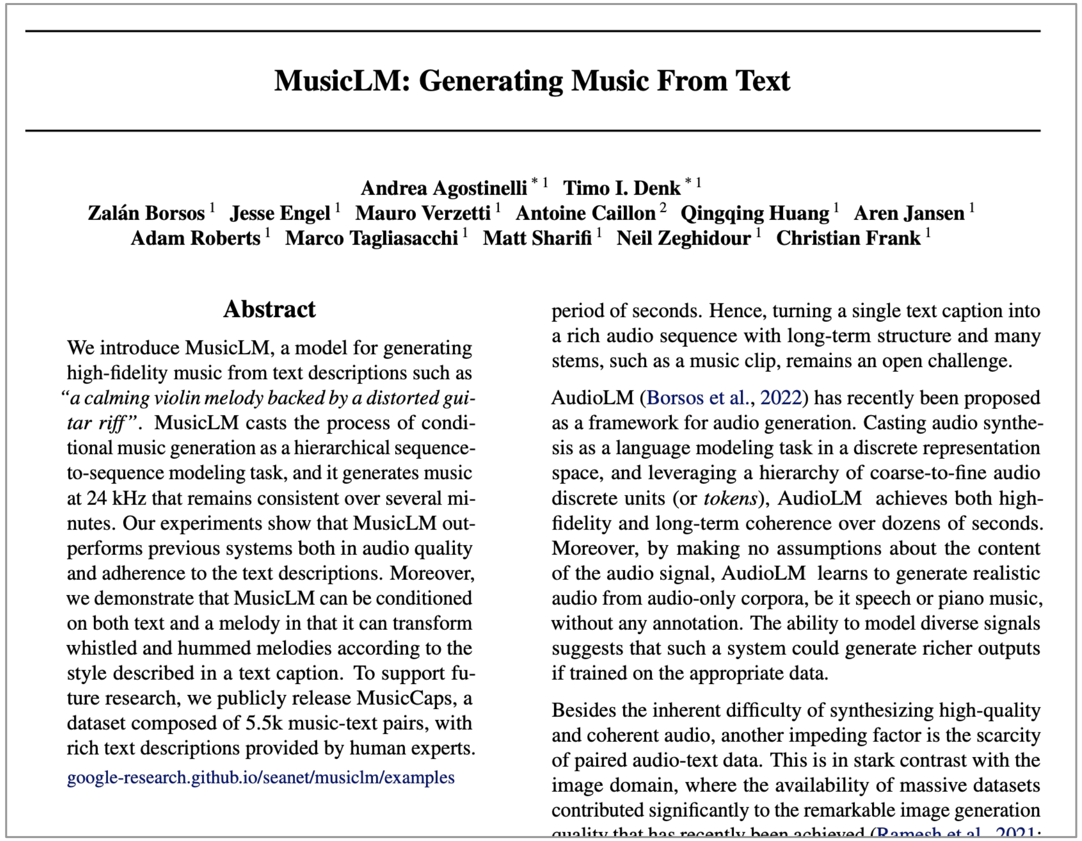
The Technics of Generative Music ML

- It’s not like sampling*
- Recognise the technical achievement
- But aesthetic & axiological critique

* *sampling* is an overloaded word in this context, which makes life hard! Here, I mean what musicians do with samplers. Generative ML systems, meanwhile, work on the basis of *drawing samples from modelled probability distributions*.

A Generative music machine learning system

- We’ll look at Agostinelli et al (2023) as an example
- Generative music ML systems typically an array of interconnected models
- Representation of audio (and therefore provenance) highly abstracted, distributed



MusicLM: Generating Music From Text

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Abstract

We introduce MusicLM, a model for generating high-fidelity music from text descriptions such as *“a calming violin melody backed by a distorted guitar riff”*. MusicLM casts the process of conditional music generation as a hierarchical sequence-to-sequence modeling task, and it generates music at 24 kHz that remains consistent over several minutes. Our experiments show that MusicLM outperforms previous systems both in audio quality and adherence to the text descriptions. Moreover, we demonstrate that MusicLM can be conditioned on both text and a melody in that it can transform whistled and hummed melodies according to the style described in a text caption. To support future research, we publicly release MusicCaps, a dataset composed of 5.5k music-text pairs, with rich text descriptions provided by human experts.
google-research.github.io/seanet/musiclm/examples

period of seconds. Hence, turning a single text caption into a rich audio sequence with long-term structure and many stems, such as a music clip, remains an open challenge.

AudioLM (Borsos et al., 2022) has recently been proposed as a framework for audio generation. Casting audio synthesis as a language modeling task in a discrete representation space, and leveraging a hierarchy of coarse-to-fine audio discrete units (or *tokens*), AudioLM achieves both high-fidelity and long-term coherence over dozens of seconds. Moreover, by making no assumptions about the content of the audio signal, AudioLM learns to generate realistic audio from audio-only corpora, be it speech or piano music, without any annotation. The ability to model diverse signals suggests that such a system could generate richer outputs if trained on the appropriate data.

Besides the inherent difficulty of synthesizing high-quality and coherent audio, another impeding factor is the scarcity of paired audio-text data. This is in stark contrast with the image domain, where the availability of massive datasets contributed significantly to the remarkable image generation quality that has recently been achieved (Ramesh et al., 2021;

One system, many models...

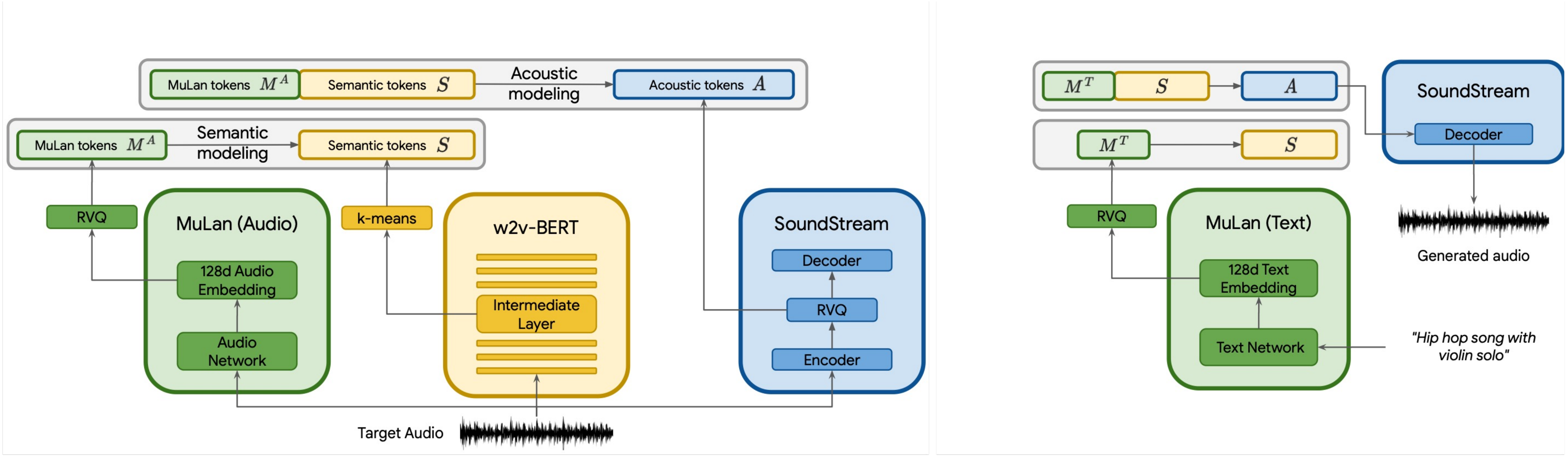


Figure 2. Left: During training we extract the MuLan audio tokens, semantic tokens, and acoustic tokens from the *audio-only* training set. In the semantic modeling stage, we predict semantic tokens using MuLan audio tokens as conditioning. In the subsequent acoustic modeling stage, we predict acoustic tokens, given both MuLan audio tokens and semantic tokens. Each stage is modeled as a sequence-to-sequence task using decoder-only Transformers. Right: During inference, we use MuLan text tokens computed from the text prompt as conditioning signal and convert the generated audio tokens to waveforms using the SoundStream decoder.

7 (and a half) models

SoundStream	1. A neural codec: i.e. a neural network version of MP3 / AAC. ‘acoustic’ tokens taken from internal representation	Representation
w2v-BERT + k-means	2 & 3. Learn representation of ‘long term’ patterns in data. w2v-BERT originally for speech recognition, repurposed here for music. K-means (clustering) used for discretizing w2v-BERT representation into ‘semantic’ tokens	
MuLan + RVQ	4 & 5. The ‘UI’ that maps text to audio. ‘RVQ’ used to produce discrete representation	
Semantic modelling	6. Generate longer term sequences, using MuLan tokens (4 & 5) to condition the generation of ‘semantic’ tokens (2 &3)	Generation
Acoustic modelling	7 & 7.5. Generate sequences that issue audio, using MuLan tokens and semantic tokens to condition the generation of acoustic tokens (from 1). Split into two stages of ‘coarse’ and ‘fine’	

7 (and a half) models

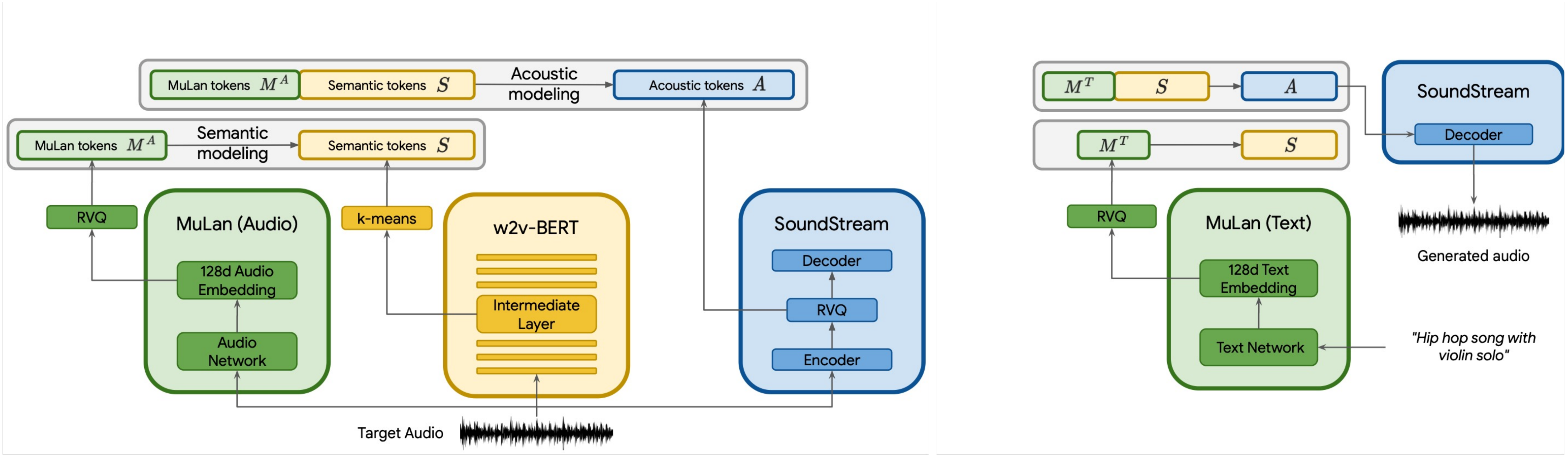
Intensifies what we covered in relation to genre:
- Radical reduction based on large scale induction
- Acute need here for significant amounts of compression

Representation

Again, from genre talk:
- Much effort needed to translate techniques from language modelling (token sequences)
- Because music doesn't have *easy* equivalents to words / sentences ('tokens')

Generation

One system, many models (and types of model; and datasets)



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One system, many models (and types of model; and datasets)

- Diverse types of model: neural networks / ML models aren't just one sort of thing
- Training is *mostly unsupervised*: only MuLan trains on labelled audio, and that came pre-trained
- Representations (SoundStream and w2v-BERT) trained on 8,000 hours of unlabelled audio from Free Music Archive
- Generators trained on 280,000 hours of unlabelled audio from 🤷‍♂️
- (MuLan pre-trained on ~375,000 hours of labelled audio from "internet music videos" + AudioSet dataset)

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Dataset harvested from Creative Commons music site

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YouTube?

Large-ish research dataset from Google; not just music

Key Points

- *A* **lot** of technical effort towards
  - Producing tokens
  - Modelling temporal relations [> but NB: music theory / cognitive / computation likes to reduce music to this!!]
  - Achieving ***compression***
- **Not** like a bucket of samples (or grains)
  - Representation of audio is distributed across these models
  - Data compression *smooshes* learned statistical regularities from training examples together
  - Actual provenance is basically gone in this architecture

Key Points – as a result of the technical processes:

- *A lot* of technical effort towards
  - Pr
  - M
  - Ac
- **Not**
  - R
  - D ng
  - Actual provenance is basically gone in this architecture

> "…the integrity, coherence and boundaries of individual works, and thereby the cultural connotations saturating those works, as well as the identity and boundaries of authorship—are dissolved…"

3 interrelated dimensions of ontological change wrought by genAI

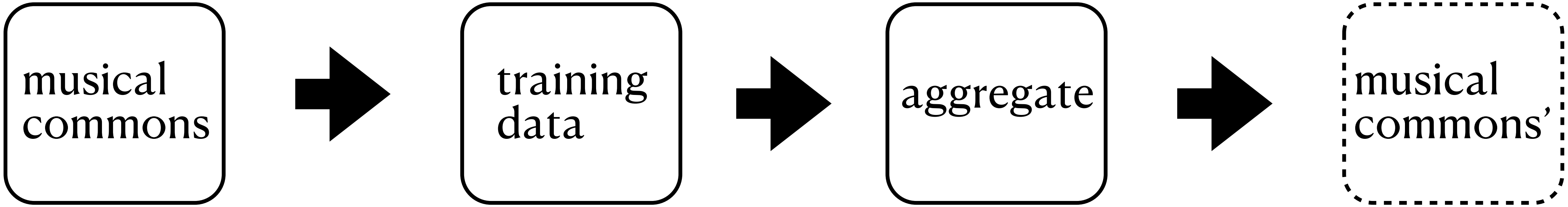
- 1) ‘Revolution’ whereby ‘fixed media’ form of recorded music gives way to infinite streams of generative and interactive music
- 2) Ontological reduction whereby AI intensifies the elevation of digitised recorded music, experienced via screens and earphones, as **the** normative experience of music – disembedded from music’s tactile and embodied, cultural and social mediation through a purification of these messy, tangible and inchoate facets rooted in offline lifeworlds > AI deepens the **solipsistic** potentials of the reduction of music to recorded sound
- 3) Materiality of AI music: clarify by comparison with earlier forms of sonic representation, montage, intertextuality associated with recorded music –
- Kenneth Goldsmith: ‘uncreative writing’ via Benjamin’s *Arcades Project* (1927-40)
- Benjamin compiled ‘hundreds of pages of notes…, a pile of shards and sketches. [The result is] a groundbreaking one-thousand-page work of appropriation…. Most of what is in the book was not written by Benjamin, rather he copied texts written by others from [a] stack of library books’. Yet ‘each entry is properly cited, and Benjamin’s “voice” inserts itself with brilliant gloss…’; ‘there is still a great deal of authorial intervention’

Ontological change 3) materiality of AI music:
GenAI musical materials: from intertextuality > aggregate

- 20th century electronic musics broadly follow Goldsmith's account of literary intertextuality: sampled sound materials are framed by a producer's or composer's **intentionality**
- Re: connotations of samples: border between acousmatic & semantic listening is blurry – aesthetic pleasures of sampling, even if sources are unknown, can be undimmed
- Sampling is, then, transitional between Benjaminian literary intertextuality and genAI
- GenAI brings a more drastic transformation of sonic materials: through the technical processes described, there's not just a weakening but an obliteration of sources and hence of Benjaminian knowing, citation-based intertextuality
- This enacts the dissolution of authorship – and attempted destruction of copyright?
- Sonic materials underlying generative AI should be conceptualised as an **aggregate** >
- Building and construction: raw materials dried and cleaned to give predictable and consistent properties, then pulverised and processed to form compound materials – or aggregate
- Processing of digitised sonic materials making up the aggregate renders them ripe for the transformational processes enacted by neural nets, detaching them from their sources – authors, musical histories – and social & cultural connotations, central to intertextuality

From aggregate to (musical) commons

ML training creates a model that corresponds to but ≠ extant musical commons



- Draws from existing musical commons
- Also adumbrates possible-but-as-yet *unrealized* commons, via de-individuation of music effected by aggregate
- Unrealized, because the resource is enclosed, monopolized, and monetized by AI companies

Conceptualizing cultural commons

- Abundant literature on management of commons, but mainly focused on natural resource commons
- Distinctive characteristics of cultural commons
  - Public good character of culture as common-pool resource (CPR): nonrivalrous and nonexcludable
  - CPR not given, but need to be produced; thus depends on ongoing reproduction of institutions, infrastructures, and cultural producers
- Distinctive characteristics of cultural commons potentialised by ML
  - Emergence of commonalities not frictionless; reductions involved in scraping, training involve ontological violence
  - Massive scale of aggregation: potential to encompass anything that can be recorded & digitized

Commons, communities, and the challenge of political organisation

- Work by Ostrom (1994) on successful management natural resource commons emphasises
  - limited size and defined boundaries;
  - clearly defined groups of (local) stakeholders; and
  - clearly defined rules governing use
- The massive, heterogeneous commons summoned by large music models defy such guidelines
- Later work by Ostrom (2010) on management of global commons (i.e., climate) emphasises instead "polycentric" approach, combination of bottom-up and top-down efforts

Four scenarios to counter AI extractivism

- What is to be done?
- Polycentric response to AI music demands broad coalition to be engaged in democratic management and decision making around commons affected/constituted by generative AI:
    - not just songwriters, recording artists, but also session musicians, instrument makers, preset designers, sound engineers, as well as listeners, critics, etc.
- But what kind of measures should such a coalition push for?

Four scenarios

1. Adaptation: hope that copyright as it currently exists provides adequate safeguard

- Problem: may protect individual creators but does nothing to address ongoing expropriation of commons

2. Redistribution: treat effects of AI on music culture as negative externality that should be regulated as such (e.g., taxation) > over

3. Regulation: beyond taxation, can use state power to levy fines, break up companies, etc. to penalize/disincentivize harmful behavior of AI companies (e.g., designating AI companies as "cultural polluters")

4. De-economise culture: limit negative impact of AI on music culture and livelihoods of creators by emancipating latter from market dependence (via UBI, decommodification of basic services, collectivisation of AI companies...)

> 4 scenarios not mutually exclusive, could be pursued simultaneously: key is to develop "non-reformist reforms" that create path dependency towards systemic change

2. Redistribution

- The problem is that the costs of the development and use of these technologies have not been adequately “priced in” to the services that commercialize them. One policy response to such market dysfunctions is to impose a tax as a way of disincentivising harmful business practices (e.g., carbon taxes)…
- The revenue generated by a hypothecated tax on the profits made from generative AI’s extraction of musical value could fund efforts to mitigate the effects of these externalities, for instance by sustaining music through support for education, skills trainings, venues and other critical infrastructure. In this way, the tax would fund just that social reproduction of the arts called for by recognizing music as culture, as the product of collective creation—as commons.

3 discussion topics (reprise!)

- 1) Copyright's informational remainder (CIR)
- 2) The aggregate
- 3) ML, music, and the commons – and alternative future models